Shopify Sales & Customer Retention Analysis

Author / Business Analyst: Project Deliverable | **Tool:** Power BI Desktop | **Dataset:** Shopify E-Commerce (March 18–24, 2025)

Total Revenue	Total Orders	Unique Customers	Avg Order Value (AOV)	Repeat Purchase Rate
$4,598,960.87	**7,431**	**4,431**	**$618.89**	**46.02%**

Stage 0: Problem Statement & Objectives

- **Goal & Stakeholders:** Provide e-commerce leadership with clear diagnostic visibility into order volume, category revenue concentration, and retention dynamics to optimize ad spend and inventory depth.
- **Key Business Questions:** (1) What is baseline revenue, order volume, and AOV? (2) Which categories drive the bulk of revenue? (3) What proportion of buyers return? (4) What is the average time between repeat checkouts? (5) How concentrated is spend among top tiers?
- **Primary Deliverable:** An interactive Power BI (.pbix) dashboard supported by this formal business analysis report.

Stage 1: Data Preparation & Star Schema Modeling

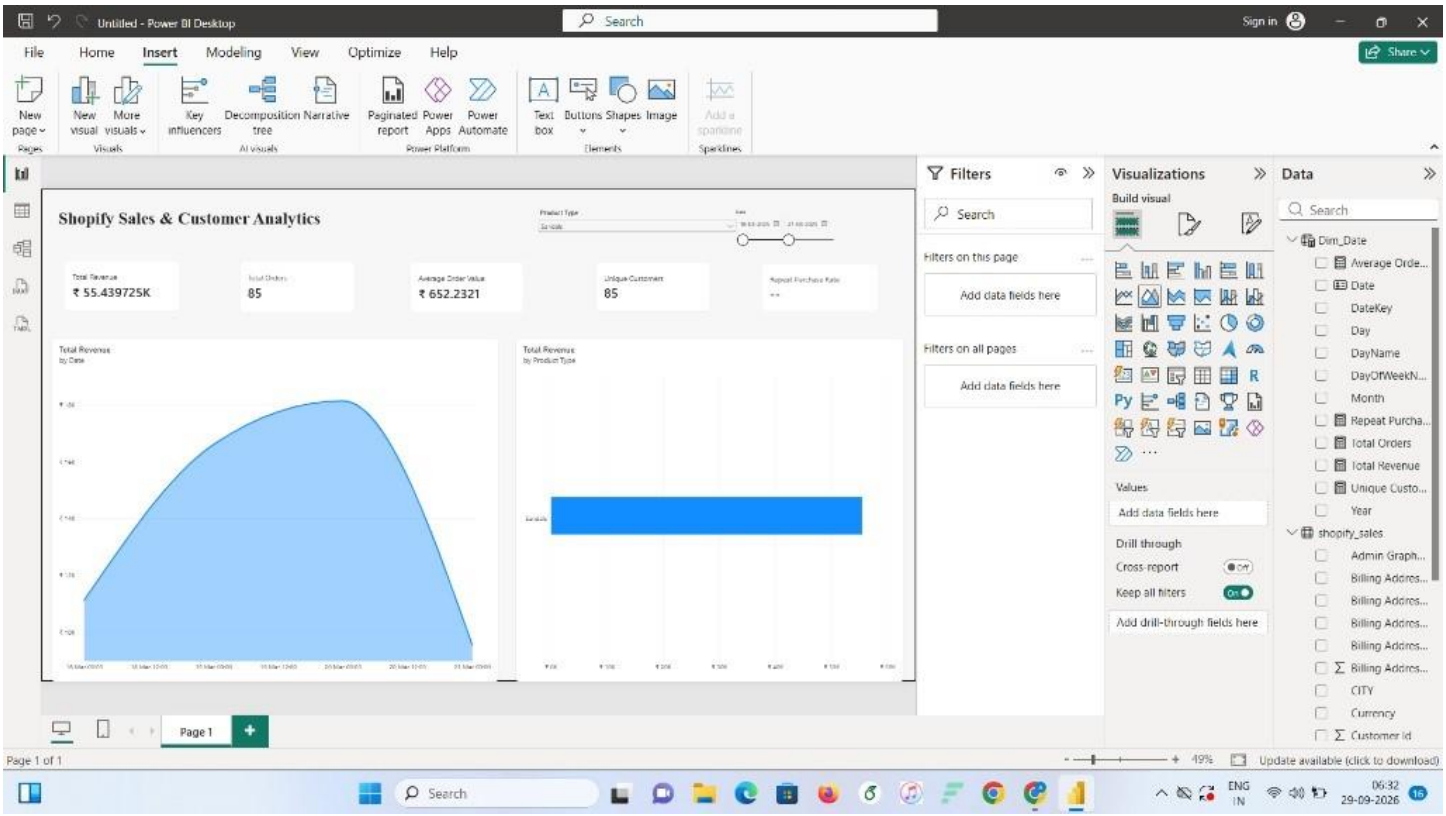
- **ETL Cleaning (Power Query):** Ingested 7,431 transaction records (19 columns). Standardized text to uppercase for `CITY`, converted numeric pricing columns to Fixed Decimal Number ($), and extracted pure Date and Time components from `Invoice Date` to ensure exact time-series matching.
- **Dedicated Calendar Dimension:** Created `Dim_Date` via DAX ranging from March 18 to March 24, 2025, marked it as official Date Table, and created a 1-to-many relationship with `shopify_sales`.

Stage 2: Transaction Performance & Core DAX Measures

Measure Name	DAX Formula	Calculated Value
Total Revenue	`SUM(shopify_sales[Total Price Usd])`	$4,598,960.87
Total Orders	`DISTINCTCOUNT(shopify_sales[Order Number])`	7,431
Average Order Value	`DIVIDE([Total Revenue], [Total Orders], 0)`	$618.89
Unique Customers	`DISTINCTCOUNT(shopify_sales[Customer Id])`	4,431
Repeat Purchase Rate	`DIVIDE(COUNTROWS(FILTER(VALUES(shopify_sales[Customer Id]), CALCULATE(DISTINCTCOUNT(shopify_sales[Order Number])) > 1)), [Unique Customers], 0)`	46.02%

Dashboard Implementation & Live Interface Screenshot

Below is the actual screenshot of the built Power BI Desktop application showing the verified Star Schema visuals, KPI callout cards, dynamic area trend curve, category horizontal bar chart, and active filter slicers:



Filter State & Verification Note:

The screenshot demonstrates active dynamic cross-filtering. When filtered to a single product line (e.g., Sandals between March 18–21), the dashboard accurately computes the exact filtered subset (85 Orders, $55.44K Revenue, $652.23 AOV). Clearing the slicer restores store-wide totals ($4.60M Revenue, 7,431 Orders, 46.02% Repeat Rate).

Stages 3 & 4: Customer Purchasing Behaviour & Retention Analysis

- **Order Frequency:** 2,392 buyers (53.98%) ordered once, while 2,039 buyers (46.02%) ordered 2 to 7 times, proving strong baseline brand affinity.
- **Spend Concentration (Pareto):** The top 10% of customers (444 buyers) generate **$1,010,240.76 (21.97%)**, and the top 20% (887 buyers) generate **$1,740,354.14 (37.84%)**.
- **Repurchase Gap Analysis:** The average latency between first and second purchase is **2.06 days (49.54 hours)** with a median of **1.82 days (43.60 hours)**.

Daily Acquisition Cohort Retention Matrix

Cohort Day	Size	Day +1	Day +2	Day +3	Day +4	Day +5	Day +6
Mar 18	945	125 (13%)	132 (14%)	151 (16%)	169 (18%)	135 (14%)	148 (16%)
Mar 19	965	158 (16%)	126 (13%)	151 (16%)	124 (13%)	150 (16%)	-
Mar 20	844	107 (13%)	127 (15%)	117 (14%)	135 (16%)	-	-
Mar 21	582	105 (18%)	70 (12%)	104 (18%)	-	-	-
Mar 22	595	87 (15%)	108 (18%)	-	-	-	-
Mar 23	412	70 (17%)	-	-	-	-	-
Mar 24	489	-	-	-	-	-	-

Stage 5: Strategic Business Recommendations

1. **Automate 48-Hour Cross-Sell Triggers:** Because repeat checkouts peak within 49.5 hours, configure transactional email and SMS flows 24–36 hours post-purchase with complementary gear (socks, shoe cleaner, insoles) to capture intent before decay.
2. **Prioritize Top Footwear Lines:** Running Shoes ($1.62M) and Tennis Shoes ($972K) account for 56.3% of total revenue. Maintain safety stock and prioritize marketing acquisition spend on these core categories.
3. **Reallocate Dead Stock Capital:** Clogs ($7.5K), Boy's ($6.9K), and Water Shoes ($2.8K) together represent under 0.4% of revenue. Liquidate underperforming SKUs and reallocate budget into high-demand footwear lines.